

Comprehensive Model Results Summary

Hedonic Price Index Analysis

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14 Models: 7 Types $\times$ 2 Time Dimensions (Quarterly/Annual)

Quarterly Models - Cumulative Jevons Indices

Model	Final Chained Jevons Log Index (log points)	Cumulative Deflation	R ² (price-change)
Traditional Jevons	-1.4231	-75.90%	N/A
Levels Hedonic	-0.6759	-49.13%	0.1279
Levels Hedonic with Lagged Errors	-0.7226	-51.45%	0.0936
Price-Change Hedonic	-1.4110	-75.61%	0.2559
Price-Change Hedonic with Lagged Errors	-1.4020	-75.39%	0.2772
Price-Change Hedonic with OLS	-1.2952	-72.62%	0.4364
Levels Hedonic with Time-Dummy Variables	-1.0399	-64.65%	0.0294 (partial R ²)

*Note: Final Chained Jevons Log Index is cumulative log change. Cumulative Deflation (%) = $100 * (\exp(\log \text{ index}) - 1)$.*

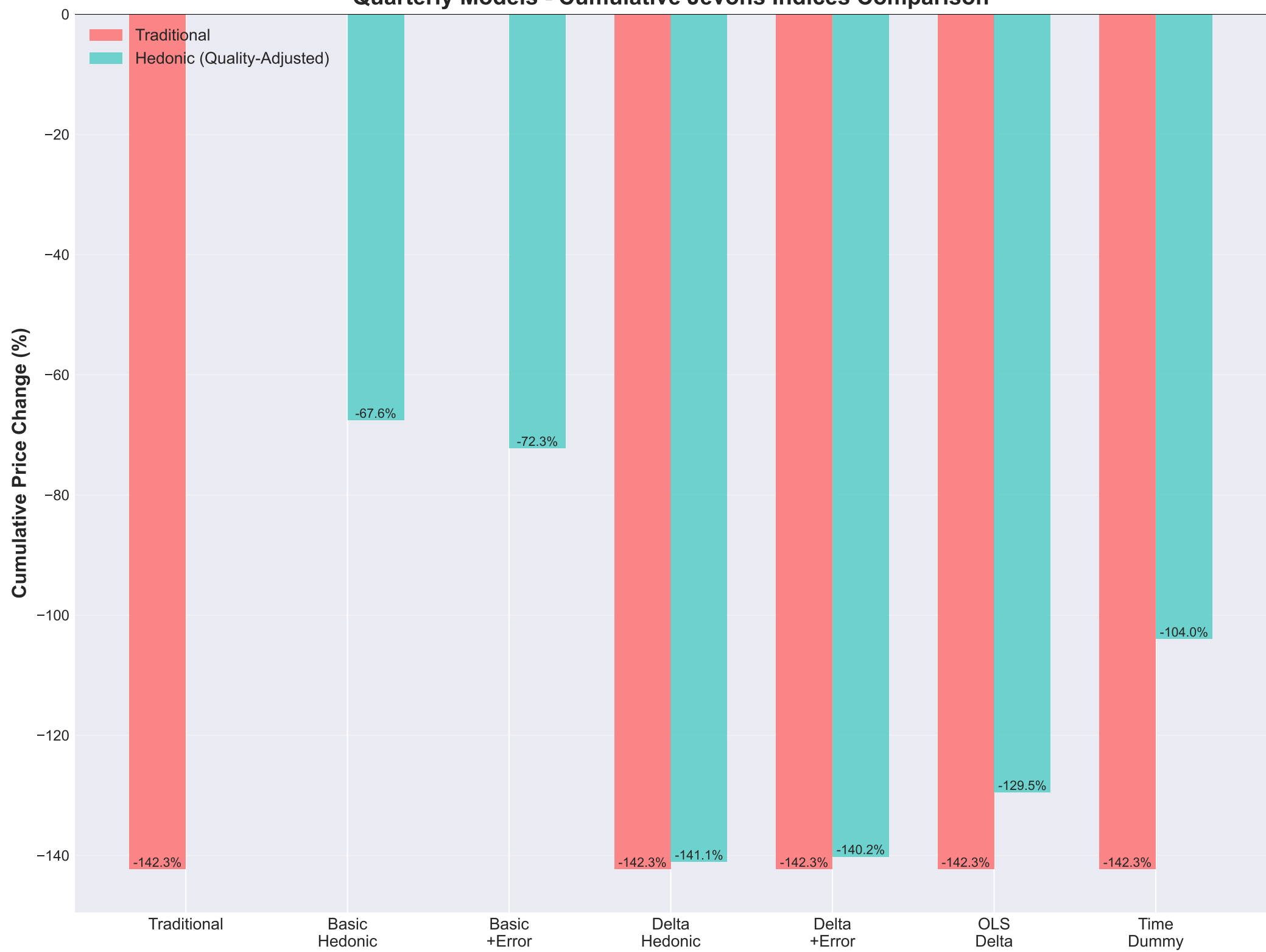
Annual Models - Cumulative Jevons Indices

All models cover 2020 to 2025 (5 periods)

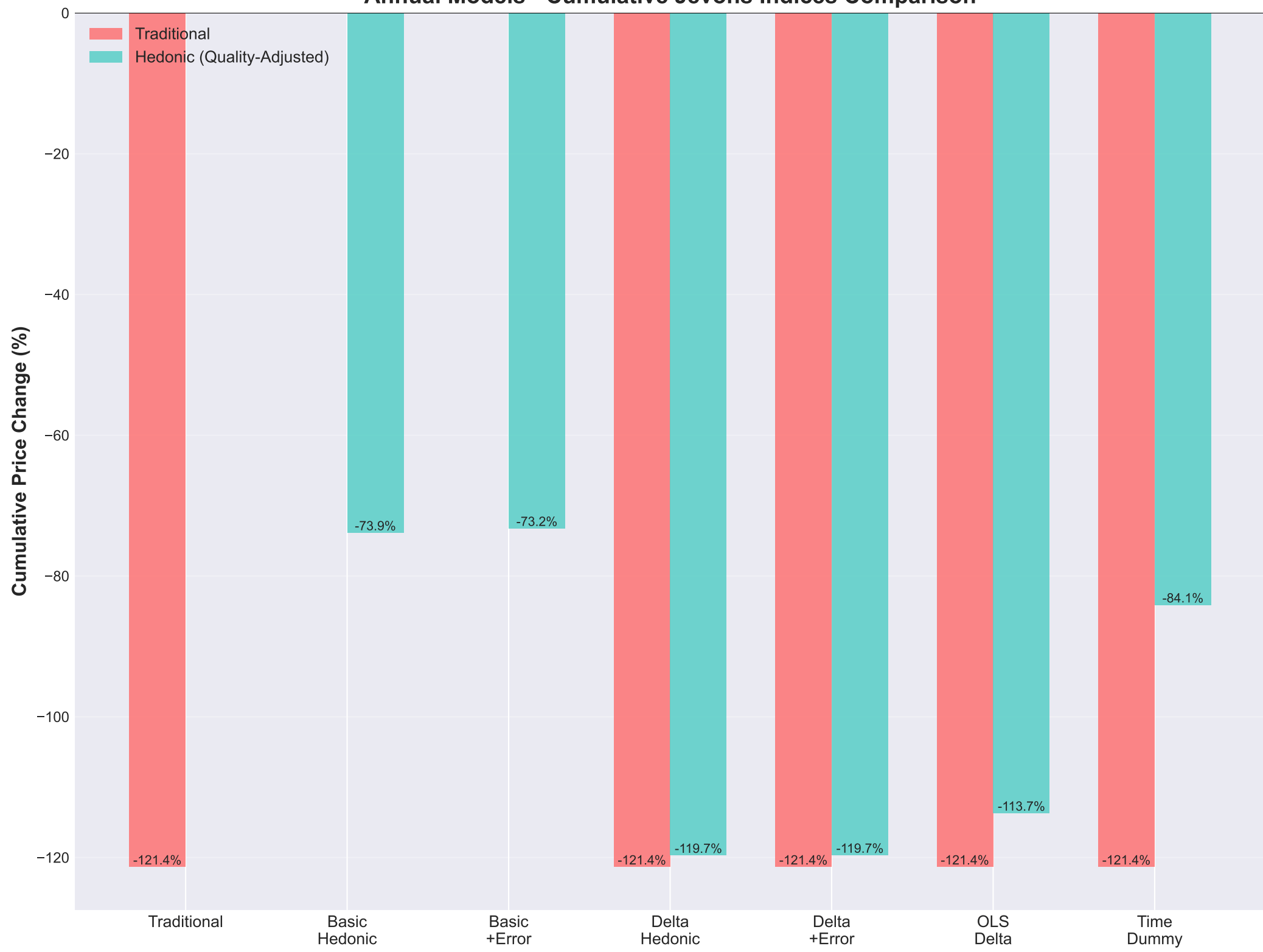
Model	Final Chained Jevons Log Index (log points)	Cumulative Deflation	R ² (price-change)
Traditional Jevons	-1.2135	-70.29%	N/A
Levels Hedonic	-0.7387	-52.22%	0.1915
Levels Hedonic with Lagged Errors	-0.7325	-51.93%	0.1012
Price-Change Hedonic	-1.1970	-69.79%	0.2498
Price-Change Hedonic with Lagged Errors	-1.1968	-69.79%	0.2623
Price-Change Hedonic with OLS	-1.1371	-67.92%	0.4226
Levels Hedonic with Time-Dummy Variables	-0.8412	-56.88%	0.0821 (partial R ²)

*Note: Final Chained Jevons Log Index is cumulative log change. Cumulative Deflation (%) = $100 * (\exp(\log \text{ index}) - 1)$.*

Quarterly Models - Cumulative Jevons Indices Comparison



Annual Models - Cumulative Jevons Indices Comparison



Model Descriptions

- **Traditional Jevons Index:**

Direct calculation using actual prices. No quality adjustment. Serves as baseline.

- **Basic Hedonic:**

Standard hedonic regression. Independent Lasso model for each period. Controls for product features.

- **Basic + Error Feature:**

Hedonic regression with previous period prediction error as additional feature. Captures time dependencies.

- **Delta Model (Lasso):**

Directly models log price differences between consecutive periods using Lasso. Uses OOF predictions to avoid mean-matching artifacts. Automatic feature selection.

- **Delta + Error Feature:**

Combines Delta model with error feature from previous period pair. Captures both price changes and time dependencies.

- **OLS Delta Model:**

Directly models log price differences using OLS (no regularization). All features included. Simpler interpretation.

- **Time Dummy Model:**

Pools consecutive periods data together. Adds time dummy variable. Single model predicts both periods. More efficient parameter sharing.